

Proof: Consider the map $A \xrightarrow{p} A$, $a \mapsto pa$, multiplication by the positive integer p . Denote the kernel, image, and cokernel of this map by ${}_pA$, pA , and A_p , respectively. The short exact sequences $0 \rightarrow {}_pA \rightarrow A \rightarrow pA \rightarrow 0$ and $0 \rightarrow pA \rightarrow A \rightarrow A_p \rightarrow 0$ give two six-term exact sequences involving $\text{Hom}(-, \mathbb{Z})$ and $\text{Ext}(-, \mathbb{Z})$. The parts of these exact sequences we need are

$$\begin{aligned} 0 \rightarrow \text{Hom}(pA, \mathbb{Z}) &\xrightarrow{\cong} \text{Hom}(A, \mathbb{Z}) \rightarrow \text{Hom}({}_pA, \mathbb{Z}) = 0 \\ \text{Hom}(pA, \mathbb{Z}) &\rightarrow \text{Ext}(A_p, \mathbb{Z}) \rightarrow \text{Ext}(A, \mathbb{Z}) \end{aligned}$$

where the term $\text{Hom}({}_pA, \mathbb{Z})$ in the first sequence is zero since ${}_pA$ is a torsion group.

Now let p be a prime, so A_p is a vector space over $\mathbb{Z}_p$. If this vector space is infinite-dimensional, it is an infinite direct sum of $\mathbb{Z}_p$'s and $\text{Ext}(A_p, \mathbb{Z})$ is the direct product of an infinite numbers of $\mathbb{Z}_p$'s, hence uncountable. Exactness of the second sequence above then implies that one of the two adjacent terms $\text{Ext}(A, \mathbb{Z})$ or $\text{Hom}(pA, \mathbb{Z}) \approx \text{Hom}(A, \mathbb{Z})$ must be uncountable, so we are done when A_p is infinite.

At the other extreme is the possibility that $A_p = 0$. This means that $A = pA$, so every element of A is divisible by p . Hence if A is nontrivial, it then contains a subgroup isomorphic to either $\mathbb{Z}[1/p]$ or $\mathbb{Z}_{p^\infty}$. We have seen that $\text{Ext}(\mathbb{Z}[1/p], \mathbb{Z}) \approx \hat{\mathbb{Z}}_p/\mathbb{Z}$ and $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) \approx \hat{\mathbb{Z}}_p$, an uncountable group in either case. As noted earlier, an inclusion $B \hookrightarrow A$ induces a surjection $\text{Ext}(A, \mathbb{Z}) \rightarrow \text{Ext}(B, \mathbb{Z})$, so it follows that $\text{Ext}(A, \mathbb{Z})$ is uncountable when $A_p = 0$ and $A \neq 0$.

The remaining case that A_p is a finite direct sum of $\mathbb{Z}_p$'s will be reduced to the case $A_p = 0$. Choose finitely many elements of A whose images in A_p are a set of generators, and let $B \subset A$ be the subgroup generated by these elements. Thus the map $B_p \rightarrow A_p$ induced by the inclusion $B \hookrightarrow A$ is surjective. The functor $A \mapsto A_p$ is the same as $A \mapsto A \otimes \mathbb{Z}_p$, so exactness of $B \rightarrow A \rightarrow A/B \rightarrow 0$ implies exactness of $B_p \rightarrow A_p \rightarrow (A/B)_p \rightarrow 0$, and hence $(A/B)_p = 0$. If A is not finitely generated, A/B is nonzero, so the preceding case implies that $\text{Ext}(A/B, \mathbb{Z})$ is uncountable. This implies that $\text{Ext}(A, \mathbb{Z})$ is also uncountable via the exact sequence $\text{Hom}(B, \mathbb{Z}) \rightarrow \text{Ext}(A/B, \mathbb{Z}) \rightarrow \text{Ext}(A, \mathbb{Z})$, since $\text{Hom}(B, \mathbb{Z})$ is finitely generated and therefore countable. $\square$

From this proposition one might conjecture that cohomology groups with $\mathbb{Z}$ coefficients are either finitely generated or uncountable.

As was explained in §3.1, the functor Ext generalizes to a sequence of functors Ext_R^n for modules over a ring R . In this generality the six-term sequences of Proposition 3F.11 become long exact sequences of Ext_R^n groups associated to short exact sequences of R -modules. These are derived in a similar fashion, by constructing short exact sequences of free resolutions. There are also analogous long exact sequences for the functors Tor_n^R , specializing to six-term sequences when $R = \mathbb{Z}$. These six-term sequences are perhaps less useful than their Ext analogs, however, since Tor is

less mysterious than Ext for nonfinitely generated groups, as it commutes with direct limits, according to an exercise for §3.A.

Exercises

1. Given maps $f_i: X_i \rightarrow X_{i+1}$ for integers $i < 0$, show that the ‘reverse mapping telescope’ obtained by glueing together the mapping cylinders of the f_i ’s in the obvious way deformation retracts onto X_0 . Similarly, if maps $f_i: X_i \rightarrow X_{i+1}$ are given for all $i \in \mathbb{Z}$, show that the resulting ‘double mapping telescope’ deformation retracts onto any of the ordinary mapping telescopes contained in it, the union of the mapping cylinders of the f_i ’s for i greater than a given number n .
2. Show that $\varprojlim^1 G_i = 0$ if the sequence $\cdots \rightarrow G_2 \xrightarrow{\alpha_2} G_1 \xrightarrow{\alpha_1} G_0$ satisfies the *Mittag-Leffler condition* that for each i the images of the maps $G_{i+n} \rightarrow G_i$ are independent of n for sufficiently large n .
3. Show that $\text{Ext}(A, \mathbb{Q}) = 0$ for all A . [Consider the homology with $\mathbb{Q}$ coefficients of a Moore space $M(A, n)$.]
4. An abelian group G is defined to be *divisible* if the map $G \xrightarrow{n} G$, $g \mapsto ng$, is surjective for all $n > 1$. Show that a group is divisible iff it is a quotient of a direct sum of $\mathbb{Q}$ ’s. Deduce from the previous problem that if G is divisible then $\text{Ext}(A, G) = 0$ for all A .
5. Show that $\text{Ext}(A, \mathbb{Z})$ is isomorphic to the cokernel of $\text{Hom}(A, \mathbb{Q}) \rightarrow \text{Hom}(A, \mathbb{Q}/\mathbb{Z})$, the map induced by the quotient map $\mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z}$. Use this to get another proof that $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) \approx \hat{\mathbb{Z}}_p$ for p prime.
6. Show that $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}_p) \approx \mathbb{Z}_p$.
7. Show that for a short exact sequence of abelian groups $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, a Moore space $M(C, n)$ can be realized as a quotient $M(B, n)/M(A, n)$. Applying the long exact sequence of cohomology for the pair $(M(B, n), M(A, n))$ with any coefficient group G , deduce an exact sequence

$$0 \rightarrow \text{Hom}(C, G) \rightarrow \text{Hom}(B, G) \rightarrow \text{Hom}(A, G) \rightarrow \text{Ext}(C, G) \rightarrow \text{Ext}(B, G) \rightarrow \text{Ext}(A, G) \rightarrow 0$$
8. Show that for a Moore space $M(G, n)$ the Bockstein long exact sequence in cohomology associated to the short exact sequence of coefficient groups $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ reduces to an exact sequence

$$0 \rightarrow \text{Hom}(G, A) \rightarrow \text{Hom}(G, B) \rightarrow \text{Hom}(G, C) \rightarrow \text{Ext}(G, A) \rightarrow \text{Ext}(G, B) \rightarrow \text{Ext}(G, C) \rightarrow 0$$
9. For an abelian group A let $p: A \rightarrow A$ be multiplication by p , and let ${}_pA = \text{Ker } p$, $pA = \text{Im } p$, and $A_p = \text{Coker } p$ as in the proof of Proposition 3F.12. Show that the six-term exact sequences involving $\text{Hom}(-, \mathbb{Z})$ and $\text{Ext}(-, \mathbb{Z})$ associated to the short exact sequences $0 \rightarrow {}_pA \rightarrow A \rightarrow pA \rightarrow 0$ and $0 \rightarrow pA \rightarrow A \rightarrow A_p \rightarrow 0$ can be spliced together to yield the exact sequence across the top of the following diagram